

Homework #4 — Multiple Linear Regression (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Sunday, 10/11 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as two separate .pdf files.
 - Answers to all textbook problems as a .pdf file in the HW4: Textbook Problems dropbox on Gradescope.
 - Knitted Quarto document as a .pdf to Insurance problem in the HW4: Insurance dropbox on Gradescope.
- Responses can be hand-written or typed for the textbook problems. You will write code with RStudio for Part II.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

We will now focus on multiple linear regression in this homework, where more of the focus will be on items such as interpretations and variable selection via adjusted R^2 , while other items such as EDA, diagnostics, and prediction will still be discussed. This will be done with textbook problems first, followed by the use of R for an insurance dataset.

Note that these concepts will come up throughout the entire week, so the homework will be done as the week goes on.

Part I: Textbook Problems (10 pts)

Complete the following problems:

1. Exercises from Chapter 8: Q3, Q8, Q12, Q14, Q16

Part II: Insurance (5 pts)

We will focus on an insurance dataset, which has 1,338 observations with 7 variables and contains demographic information and costs regarding medical patients. More information about the dataset [here](#).

The dataset can be found beneath the link to his homework assignment on my website.

We will be using R to work with the data. For this question, you will turn in a separate, knitted Quarto document with code and output related to fitting a multiple linear regression model. Follow all instructions below.

Create a new Quarto document. Title the document “HW3, Part II” and add your name in the Name box. Download the dataset from the website and make sure that your Quarto document and dataset are **in the same folder**. Use the following code to bring the dataset into your R environment:

- `library(dplyr)`
- `library(ggplot2)`
- `insurance <- read.csv("insurance.csv", header = T)`

Answer the following questions below using R code:

- (a) Pick three predictors to use from the dataset, with one of the three being a categorical variable.
 - For the categorical variable, use the `factor()` function to convert the variable into a factor class with appropriate levels.
- (b) For each predictor, create an appropriate plot comparing the predictor to the response variable, charges. Describe the relationship between the predictor and response for each plot in 1-2 sentences.
- (c) Fit the model as specified in part (a). Write/type out the least squares line afterward.

- (d) Interpret each of the β coefficients associated with your predictors. Then, determine if that predictor is significant.
- (e) Use `plot(model, 1)` to show the residuals vs. fitted plot, with `model` being your stored fit. Describe what you see, and if there are any concerns.
- (f) Perform either a forward or backward selection procedure by fitting multiple candidate models until you cannot go any further. Discuss how much improvement you see between your original model and refined model.